JEPPIAAR
ENGINEERING COLLEGE

# DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

CS3811- Project Work
Domain: Artificial Intelligence

# AI-Driven Frustration Detection for Smarter Code Reviews


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# Literature Survey

- **Keystroke Dynamics:** Typing speed, pauses, and backspace frequency reveal cognitive load, aiding stress detection.
- **Facial Emotion Recognition:** CNN-based models classify emotions like anger, sadness, and fear that correlate with frustration.
- **Multimodal Fusion:** Combining behavioral and visual signals with weighted fusion improves real-time accuracy.
- **Workplace Stress Monitoring:** Existing systems track productivity but ignore emotional state and adaptive support.
- **Research Gap:** No IDE-level solution integrates real-time frustration detection with gamified emotional reset for developers.

# Proposed System

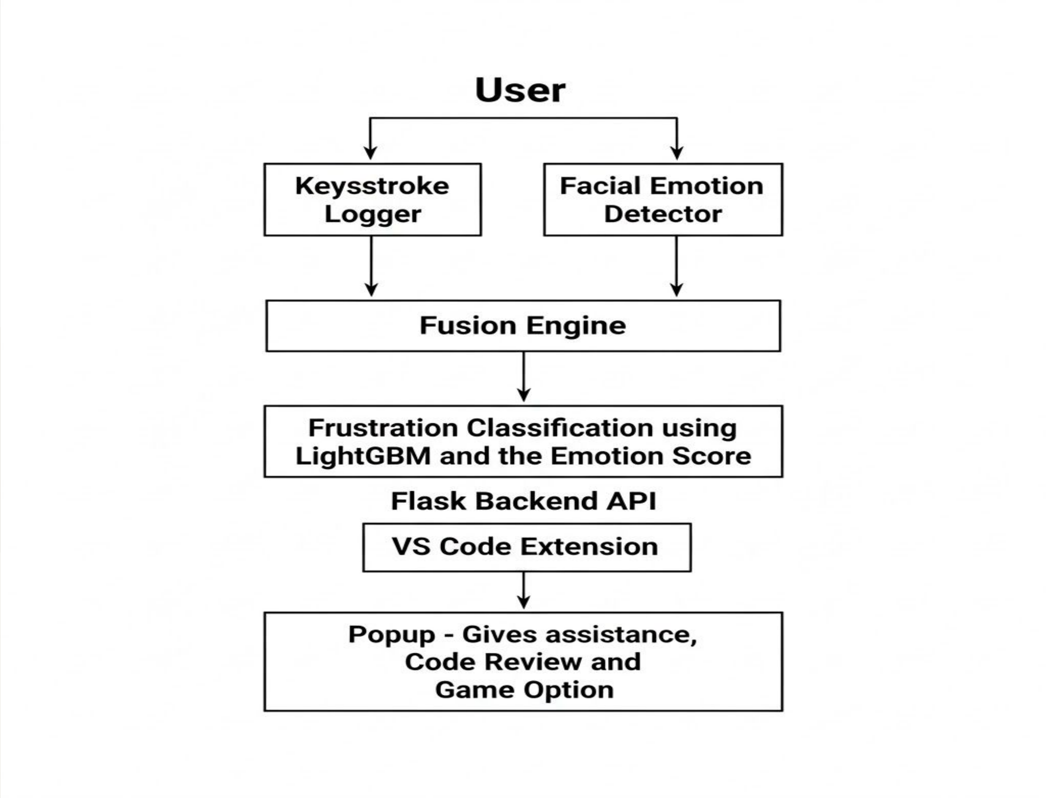
**Objective:** Detect developer frustration in real-time using behavioral and facial signals, delivering adaptive support in VS Code.

**Features:**

- Keystroke dynamics & facial emotion recognition
- LightGBM-based classification with multimodal fusion
- Integrated VS Code assistant
- Gamified reset tool

**Outcome:** Improved well-being, reduced debugging stress, and enhanced productivity awareness.

# System Architecture Diagram

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# Module Details

- Data Collection & Labeling Module
- Feature Engineering & Model Training Module
- Backend API Development Module
- VS Code Extension Development Module
- Frontend Dashboard Development Module
- System Integration & Testing Module
- Deployment & Packaging Module

# Implementation Details

- **Backend:** Python with Flask, LightGBM for classification, OpenCV & DeepFace for emotion recognition.
- **Frontend:** VS Code Extension built in TypeScript, using Webview API and Axios for API communication.
- **ML Model:** Window-based feature extraction, label encoding, stratified train-test split, saved via joblib.
- **Dataset:** Session-based CSV logging, labeled as *normal* or *frustrated*.

# Current Status

- **Completed:** Core pipeline including keystroke data collection, feature extraction, LightGBM model training, facial emotion detection, fusion engine, Flask API bridge, and VS Code prototype.
- **In Progress:** UI refinement for Assistance, integrated testing inside VS Code, and accuracy evaluation with larger datasets.
- **Planned:** Code auto-review integration, enhanced gamification features, and a team emotional dashboard for collaborative insights.